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Experimental and kinetic modeling investigation on ethylcyclohexane low-temperature oxidation in a jet-stirred reactor

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A B S T R A C T

In this work, the oxidation of ethylcyclohexane was studied in a jet-stirred reactor at 780 Torr, 480–780 K and equivalence ratios of 0.5, 1.0 and 2.0. Synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) was used for the detection of oxidation products, including a series of reactive intermediates such as cycloalkylhydroperoxides, keto-hydroperoxides, alkenal-hydroperoxides and highly oxygenated molecules. Quantum chemistry calculations were performed to obtain ionization energies for the identification of some important intermediates and energy barriers of several important pathways. On the other hand, this work presents the first efforts on developing a low-temperature oxidation model of ethylcyclohexane. The present model can reasonably capture the low-temperature oxidation reactivities and negative temperature coefficient behaviors observed in both present and previous experimental work of ethylcyclohexane oxidation. Modeling analyses were performed to provide insight into the low-temperature oxidation chemistry of ethylcyclohexane. The two-stage O_2 addition mechanism is concluded to dominate the chain-branching process in the low-temperature region. The concerted elimination reactions of cycloalkylperoxy radical and “formally direct” chemically activated reactions of cycloalkyl+O_2 result in cycloalkenes and HO_2 formation at the negative temperature coefficient region and serve as main chain-termination pathways. Compared with smaller cycloalkanes like cyclohexane and methylcyclohexane, the ethyl sidechain structure in ethylcyclohexane reduces the energy barriers of cycloalkylperoxy radical isomerization and facilitates the formation of keto-hydroperoxides which leads to more pronounced low-temperature oxidation reactivity of ethylcyclohexane.

1. Introduction

Low-temperature (low-T) oxidation of transportation fuels plays important roles in engine combustion, such as engine knock [1] and low-T combustion techniques in advanced engines [2,3]. In these cases, low-T oxidation chemistry serves as the driven force and is strongly affected by different structural features of major fuel components including alkanes, cycloalkanes and aromatics. Among the three major component families, cycloalkanes have a total composition of 15–30% in conventional transportation fuels [4,5], while the maximum composition of cycloalkanes could reach up to almost 100% in unconventional transportation fuels such as oil-sand derived fuels [6], coal-based synthetic fuels [7] and hydrogenated bio-oils [8]. Furthermore, previous studies suggested that cycloalkanes can be used as octane boosters [9] to permit the tailoring of fuel anti-knock characteristics. However, compared with alkanes, investigations on the low-T

List of abbreviations: AnHP, alkenal-hydroperoxide; CEHP, cyclic ether hydroperoxide; ECH, ethylcyclohexane; high-T, high-temperature; IDT, ignition delay time; JSR, jet stirred reactor; KHP, keto-hydroperoxide; MCH, methylcyclohexane; OFDHP, olefinic dihydroperoxide; $OOP(OOH)_2$, dihydroperoxycycloalkyl peroxy radical; PICS, photoionization cross section; $P(OOH)_2$, dihydroperoxycycloalkyl radical; RCM, rapid compression machine; ROOH, cycloalkylhydroperoxide; QOOH, hydroperoxycycloalkyl radical; SVUV-PIMS, synchrotron vacuum ultraviolet photoionization mass spectrometry; $T(OOH)_3$, trihydroperoxycycloalkyl radical; CEDHP, cyclic ether dihydroperoxide; CHX, cyclohexane; GC–MS, gas chromatography combined with mass spectrometry; HOM, highly oxygenated molecule; IE, ionization energy; KDHP, keto-dihydroperoxide; low-T, low-temperature; NTC, negative temperature coefficient; OFHP, olefinic hydroperoxide; OOQOOH, hydroperoxycycloalkyl peroxy radical; PIE, photoionization efficiency; R, cycloalkyl radical; ROO, cycloalkylperoxy radical; ROP, rate of production; ST, shock tube; TS, transition state.

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oxidation chemistry of cycloalkanes are still very limited, especially for large alkylcyclohexanes with C_2 and longer sidechain structures which widely exist in conventional and unconventional transportation fuels [10,11]. Among large alkylcyclohexanes, ethylcyclohexane (ECH) is a typical one in transportation fuels and can be used as a representative cycloalkane component in kerosene surrogate fuels, as well as a model molecule for large cycloalkane components in transportation fuels [12].

A number of studies have been conducted on the high-temperature (high-T) pyrolysis and oxidation of ECH, such as measurements of global combustion parameters including ignition delay times (IDT) [13–15] and laminar burning velocities [16,17] and species profiles in flow reactor pyrolysis and laminar premixed flames [18]. To our knowledge, there are only two previous experimental studies of ECH oxidation under low-T conditions (generally below 800 K). Husson et al. [19] detected the low-T oxidation products of ECH in a jet stirred reactor (JSR) at 780 Torr and equivalence ratios (ϕ) of 0.25, 1.0 and 2.0 using gas chromatography combined with mass spectrometry (GC–MS), including cyclic ethers, aldehydes, cycloalkenes and small decomposition products. Low-T reactivity and negative temperature coefficient (NTC) behavior were observed. Besides, Kang et al. [20] investigated the ignition process of ECH in a modified cooperative fuel research engine and performed conformational analyses to explain the different reactivities between ECH and two dimethylcyclohexanes. No measurement on reactive intermediates, such as cycloalkylhydroperoxides (ROOHs), keto-hydroperoxides (KHPs) and highly oxygenated molecules (HOMs) which play crucial roles in low-T chain-branching process, was performed for low-T oxidation of ECH. Besides experimental investigations, theoretical calculations on reaction pathways and rate constants are crucial for kinetic model development. However, to our knowledge, there are only a few theoretical studies on low-T oxidation reactions of ECH [21,22] compared with cyclohexane (CHX) [2,3,23–27] and methylcyclohexane (MCH) [21,27–30]. Similar to the situation of experimental studies, there are several kinetic models developed for high-T combustion of ECH, such as the model of Wang et al. [18] and the JetSurF 2.0 model [31], while no low-T oxidation model was developed for ECH. The high-T models were found incapable to predict the low-T oxidation behaviors of ECH, especially the complete absence of NTC region in simulation [18,22,31].

In this work, the low-T oxidation of ECH at 780 Torr was investigated in a JSR under lean, stoichiometric and rich conditions. Synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS) [32–34] was used to detect oxidation products, especially reactive intermediates like ROOH, alkenal-hydroperoxide (AnHP), KHP and HOM, with the assistance of theoretical calculations of ionization energies (IEs) of possible intermediates and energy barriers of several key pathways. Furthermore, a kinetic model was developed to predict fuel consumption and product formation under the investigated oxidation conditions. The model was also validated against the species profiles measured in previous JSR oxidation of ECH [19]. Modeling analyses including the rate of production (ROP) analysis and sensitivity analysis were performed to provide insight into the chemistry in fuel consumption and low-T chain-branching processes of ECH.

2. Experimental and theoretical methods

2.1. Jet-stirred reactor oxidation

The experiments were performed at National Synchrotron Radiation Laboratory in Hefei, China. Detailed descriptions of the VUV beamline and JSR apparatus have been introduced elsewhere [32,35–37]. The schematic diagram of the synchrotron JSR apparatus can be found in Fig. S1 in the *Supplementary Material*. The quartz JSR used in this work is very similar to that reported by Olivier and Dayma [38] which allows a perfect mixing inside the reactor through the turbulent jets generated from the nozzles [39]. The reactor has a volume of 102 cm^3 and is fused with a quartz sampling nozzle which has a ~80 μm orifice on the tip. Liquid fuel (ECH, 99.8% purity) was vaporized in an electrically heated vaporizer with argon (99.9999% purity) as the carrier gas, mixed with oxygen (99.9999% purity) at the vaporizer outlet, then entered the JSR. The fuel mole fractions in inlet mixtures were set to be 1% to avoid large chemical heat release. No unsteady phenomena, such as oscillatory cool flames or multi-stage ignitions, were observed in this work. The experiments were performed at 780 Torr and 480–780 K. The reaction temperature in the reactor was measured by a K-type thermocouple mounted in the center of the reactor with $\pm$2 K accuracy. The average residence time was maintained at 2 $\pm$ 0.03 s, while the equivalence ratios were selected at 0.5, 1.0 and 2.0. The identification and mole fraction evaluation of intermediates were based on previously reported methods [40–43]. The photoionization cross sections (PICSs) were mainly referred to our online database [44]. For major products, such as CH_4, CO, H_2O and CO_2, the uncertainties of measured mole fractions are within $\pm$20%. For intermediates with known PICSs, the uncertainties are within $\pm$50%, while for CH_3OOH with calculated PICS [45] and intermediates with estimated PICSs, the uncertainties are estimated to be a factor of 2.

2.2. Theoretical calculation

To identify the role of ethyl sidechain on the low-T oxidation process, energy barriers of several isomerization pathways were calculated by using the complete basis set methodology (CBS-QB3) which is less computationally expensive and has been successfully applied in many cycloalkane reaction systems [3,21,22,46]. Detailed descriptions of CBS-QB3 method can be found in previous studies [3,47,48] which estimated the energy uncertainty to be less than 1.5 kcal/mol. In order to identify oxidation intermediates, especially hydroperoxides, the IEs of possible structures were calculated. The most stable structure [18,46], i.e. the cyclohexane ring in chair conformation and the ethyl group at equatorial position, was selected as the initial structure. Furthermore, as suggested by Yang et al. [27], only axial peroxy or hydroperoxy group was considered in ROO, hydroperoxycycloalkyl radical (QOOH), hydroperoxycycloalkyl peroxy radical (OOQOOH) and dihydroperoxycycloalkyl radical ($P(OOH)_2$) due to the existence of cyclohexane ring. CBS-QB3 method was used to obtain the energies of neutral molecules and ions based on previous studies [47–49]. For small molecules, the uncertainties of calculated IEs were estimated to be within 0.05 eV according to previous studies [49–51]. For large molecules, especially HOMs, there are many conformers with similar energies for neutral molecules and slightly different energies for ions according to our calculated results. Previous studies revealed that the existence of conformers might lead to higher uncertainties than specified [49,52,53]. Thus, the uncertainties of calculated IEs were estimated to be within 0.15 eV for large molecules with conformers.

3. Kinetic modeling

In this work, a kinetic model of ECH oxidation was developed based on both recent progress in the low-T models of hydrocarbons and application of the reaction class concept. The AramcoMech 2.0 model [54-59] was adopted as the core C1-C4 reaction mechanism since it contains comprehensive low-T and high-T oxidation sub-mechanisms. The high-T oxidation mechanism of ECH was taken from our previous high-T ECH model [12] which was

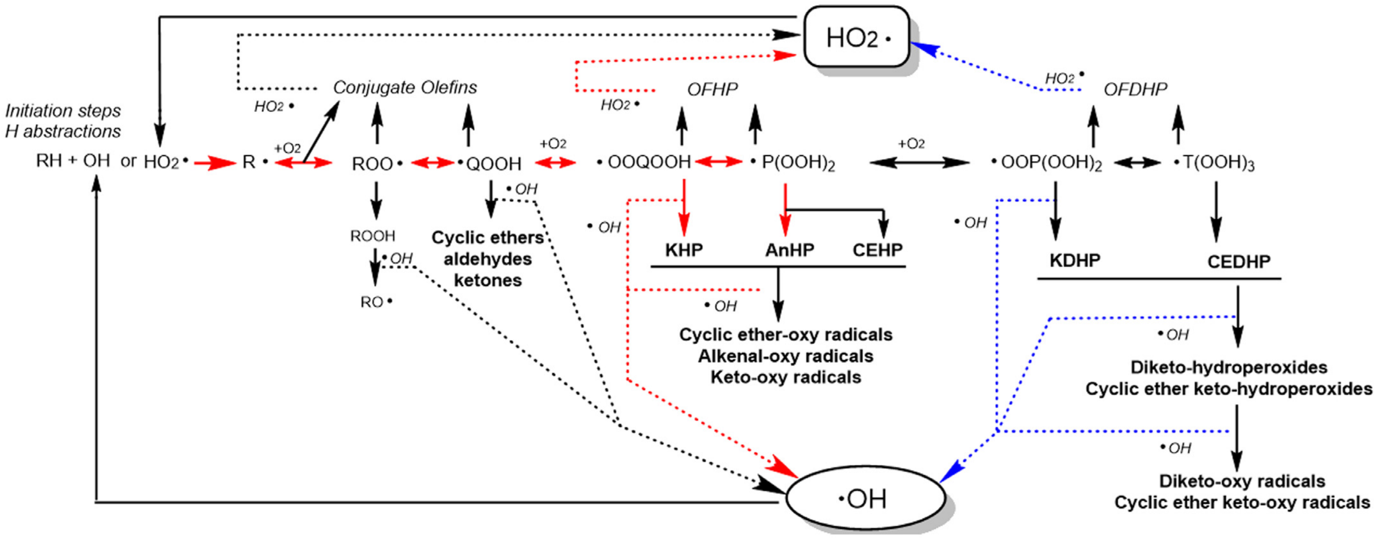
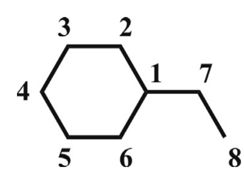


Fig. 1. Scheme of low-T oxidation reaction classes of alkylcycloalkane. Note that in this work abbreviations are only used for intermediates appeared in this scheme which have the same number of carbon atoms as the fuel. The main chain-branching pathways are highlighted in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

validated against species profiles in flow reactor pyrolysis, laminar premixed flames [16] and JSR oxidation [19], IDTs in shock tube (ST) [13] and laminar burning velocities [60] under various pressures and equivalence ratios. The low-T reaction classes of ECH were referred to a recently published MCH model [61], which was developed based on the low-T reaction classes proposed by Sarathy et al. [62], as well as the alternative isomerization, concerted elimination starting from OOQOOH radicals and the third oxygen addition to $P(OOH)_2$. In addition, some pathways related to AnHPs were proposed as new consumption pathways of $P(OOH)_2$. Figure 1 shows the scheme of the low-T oxidation reaction classes of alkylcycloalkanes. Table S1 in the *Supplementary Material* summarizes both high-T and low-T reaction classes of ECH in the present model, while Table S2 in the *Supplementary Material* provides the sources of rate constants for the reaction classes of ECH in the present model. The chemical structures of species in the ECH sub-mechanism are shown in Table S3 in the *Supplementary Material*. For species without available thermodynamic data, the THERGAS code [63] was used to estimate their thermodynamic data. The present model includes 1030 species and 4228 reactions. The simulation was performed with the Perfectly Stirred Reactor module in the Chemkin-Pro software [64] using fixed gas temperature and transition solver settings [61,65]. The reaction mechanism and thermodynamic data files of the present model can also be found in the *Supplementary Material*. The construction of the ECH sub-mechanism is briefly introduced below, which is mainly focused on the chain-branching process.

3.1. ECH decomposition and first O_2 addition

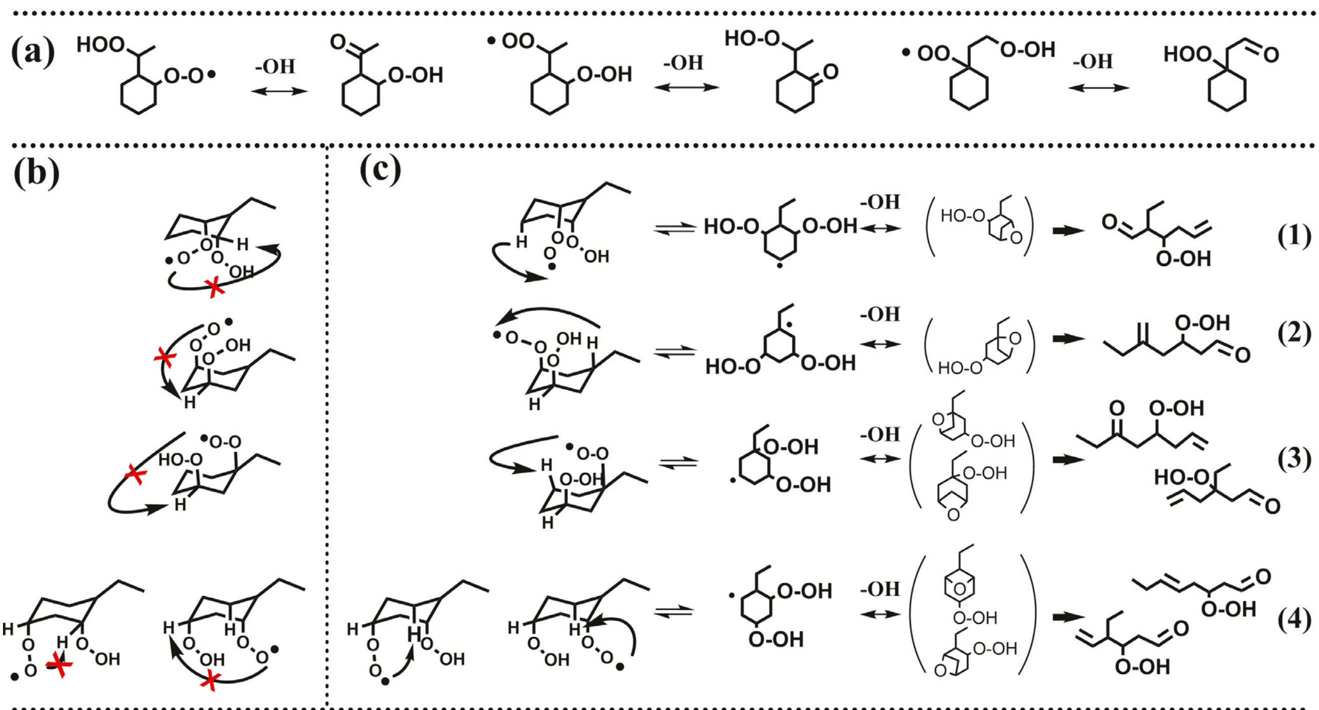
As summarized by Battin–Leclerc et al. [11,47], the first several steps in the low-T oxidation of ECH are similar to those of alkanes. Scheme 1 shows the molecular structure of ECH with eight carbon sites defined from 1 to 8. It should be noted that carbon sites 2 and 6 are at symmetric positions, so are the carbon sites 3 and 5. There are six C_8H_{15} radicals (R) produced from H-abstraction, i.e. ECHR1, ECHR2, ECHR3, ECHR4, ECHR7 and ECHR8, in which the numbers denote corresponding carbon sites. Sivaramakrishnan and Michael measured the global rate constants of reactions between OH and alkanes [66], cycloalkanes and methylcycloalkanes [67]. Subsequently, Sarathy et al. [62] proposed the reaction rate



Scheme 1. Molecular structure of ECH.

rules for site-specific H-abstraction reactions of 2-methylalkanes by OH, which also were applied to the H-abstraction reactions of MCH by OH in the work of Weber et al. [30]. In present model, the rate constants of H-abstraction reactions by OH at the ring sites were referred to similar reactions of MCH [30], while those at the sidechain sites were referred to the similar reactions of alkanes [62]. The H-abstraction by HO_2 is another important type of H-abstraction at intermediate temperatures. However, there are very few studies on this reaction type. In the present model, the rate constants of H-abstraction reactions at the ring sites by HO_2 were referred to the measured rate constants of similar reactions of CHX at 673–773 K [68], while the rate constants of those at carbon sites 1, 7 and 8 were referred to the calculated results of similar reactions of C_2–C_4 alkanes by Aguilera-Iparraguirre et al. [69].

As shown in Fig. 1, the first O_2 addition ($R + O_2$=ROO) is the second step in the low-T chain-branching process, followed by the isomerization of ROO to QOOH (ROO=QOOH). Up to now, only Ning et al. [22] investigated the first O_2 addition reactions of ECH radicals at CBS-QB3 level. Figure S2 in the *Supplementary Material* shows the comparison between the six reactions of ECH radicals calculated by Ning et al. [22] and similar reactions of alkyl radicals [70] and cyclohexyl radical [24]. The calculated rate constants of Ning et al. [22] demonstrate unexpected greater decay from the tertiary carbon site radical (ECHR1) reaction to primary carbon site radical (ECHR8) reaction than that of similar reactions of alkyl radicals [70], i.e. more than three orders of magnitude versus less than one order of magnitude at 600 K. This also causes much slower secondary ring carbon site radical (ECHR2, ECHR3 and ECHR4) reactions [22] than the similar reaction of cyclohexyl radical calculated by Fernandes et al. [24]. Due to these reasons, the first O_2 addition reactions of ECHR2, ECHR3 and ECHR4 were referred to the similar reaction of cyclohexyl radical [24] in the present model,



Scheme 2. (a) KHP formation pathways from OOQOOH with OO• and OOH moieties on both ring and sidechain carbon sites via 1,5-H shift; decomposition of OOQOOH with both OO• and OOH moieties on ring carbon sites to (b) KHP and (c) AnHP based on conformational analysis.

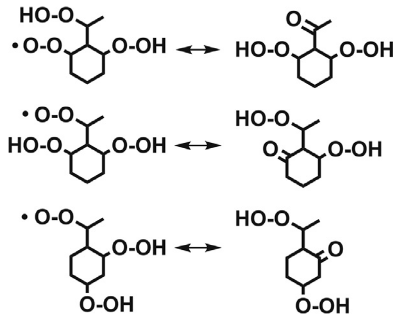
while the first O_2 addition reactions of ECHR7 and ECHR8 were referred to similar reactions of alkyl radicals [70]. Furthermore, the first O_2 addition reaction of ECHR1 was referred to that of similar reaction of cyclohexyl radical [24] with the activation energy reduced by 1 kcal/mol considering the more active tertiary carbon site than the secondary carbon sites.

For ROO isomerization, Yao et al. [21] performed a theoretical investigation on part of these reactions for ECHR1OO/ECHR2OO/ECHR3OO/ECHR7OO/ECHR8OO radicals at CBS-QB3 level, which were adopted in the present model. Besides, other ROO isomerization reactions which were not reported by Yao et al. [21] were referred to similar reactions of CHX [24], MCH [28] and alkanes [31]. For the first O_2 addition on ring carbon sites and subsequent isomerization of ECHR1OO, ECHR2OO, ECHR3OO and ECHR4OO, the competition reactions (“formally direct” chemically activated reactions of R + O_2 and concerted elimination reactions of ROO) and decomposition reactions of QOOH (QOOH=cyclic ether/alkenal+OH and QOOH=cycloalkene+HO_2) were referred to similar reactions of CHX [24] and MCH [28,29]. Furthermore, the decomposition reactions of QOOHs produced from ECHR7OO and ECHR8OO isomerization were referred to similar reactions of alkanes [71-73].

3.2. Second O_2 addition and KHP formation/decomposition

Besides decomposition, QOOH can add O_2 to form OOQOOH, which is the second O_2 addition pathway and the fourth step in the low-T chain-branching process. Recent low-T oxidation modeling work of CHX [65] and MCH [30,61] suggested that 1,5-H shift dominates the isomerization of ROO, which is in agreement with the conclusions in the theoretical work of Knepp et al. [2] and Xing et al. [28,29]. Figures S3–S5 in the *Supplementary Material* compare the calculated rate constants for the isomerization pathways of cyclohexyl peroxy radical [24], 1-methyl-cyclohex-2-yl peroxy radical [28] and 1-ethyl-cyclohex-2-yl peroxy radical [21] in literature. It demonstrates that the rate constant of 1,5-H shift is two orders of magnitude higher than those of other TS sizes. Thus in previous low-T oxidation model of MCH [61], QOOH formed from 1,5-H shift of ROO was suggested as the dominant one undergoing further O_2 addition to simplify the reaction mechanism. This strategy was also adopted in the present model, while the rate constant of the second O_2 addition was referred to that of the first O_2 addition with pre-exponential factors divided by a factor of 2 according to the theoretical calculation results in low-T alkane oxidation [74].

The isomerization and decomposition of OOQOOH producing KHP and the first OH radical is the fifth step in the low-T chain-branching process, while further decomposition of KHP producing the second OH radical and an oxygenated radical (XO) is the sixth step. According to the knowledge in low-T alkane oxidation, the rate constant of KHP formation mainly depends on the position of OO• moiety, the strength of C–H bond for the H shift and the size of TS ring in the isomerization process [6,74]. Scheme 2(a) shows the typical KHP formation pathways from OOQOOH with OO• and OOH moieties on both ring and sidechain carbon sites in the low-T oxidation of ECH. According to previous modeling work [6,30,61], the rate constant of KHP formation was referred to that of ROO=QOOH with the activation energy reduced by 3 kcal considering the weaker C–H bond [62]. In previous low-T oxidation models of MCH [30,61,75,76], researchers proposed KHP formation pathways from OOQOOH with both OO• and OOH moieties on the ring carbon sites. Due the chair structure of cyclohexane ring, the first O_2 addition and ROO isomerization on the ring carbon sites prefer to occur on axial C–H bonds, thus the formed OOQOOH have both OO• and OOH moieties on the axial positions, as shown in Scheme 2(b). It can be observed from Scheme 2(b) that in these cases, the KHP formation are inhibited due to the steric hindrance effects [77]. Therefore, in the present model, alternative pathways were proposed to describe the decomposition of OOQOOH with both OO• and OOH moieties on the ring carbon sites. As shown in Scheme 2(c), OOQOOH can shift the third axial H atom to form $P(OOH)_2$ via a 6-member TS. In the following text, 4-, 5-, 6-, 7-, 8- and 9-member will be abbreviated as 4-, 5-, 6-, 7-, 8- and 9-m, respectively. $P(OOH)_2$ can be either consumed by the third O_2 addition or decompose to 4-m CEHP (cyclic ether



Scheme 3. Proposed formation pathways of KDHPs via 1,5-H shift.

hydroperoxide). However, previous experimental [6,19,61,65] and theoretical [3] studies of low-T oxidation of cycloalkanes revealed that 4-m cyclic ether structure on the ring carbon sites is not stable. Therefore, the formed 4-m CEHP can quickly convert to AnHP, as shown in Scheme 2(c). The rate constants of these reactions were referred to the calculated rate constant of QOOH decomposition forming alkenal for CHX [24]. Furthermore, the alternative isomerization and concerted elimination of OOQOOH and decomposition reactions of $P(OOH)_2$ were also referred to the similar reactions in the first O_2 addition process.

3.3. Third O_2 addition

Recently, Wang and coworkers reported the third O_2 addition mechanism in the low-T oxidation of MCH [30,61] and *n*-butylcyclohexane [77]. Thus in the present model, the isomerization of OOQOOH to $P(OOH)_2$ and subsequent third O_2 addition to $P(OOH)_2$ were also incorporated. Similar to the second O_2 addition, recent low-T oxidation modeling work of CHX [65] and MCH [28–30, 61] also suggested the dominance of 1,5-H shift in the isomerization of OOQOOH. Therefore, in order to simplify the reaction mechanism, $P(OOH)_2$ formed from 1,5-H shift of OOQOOH was selected as the one undergoing further third O_2 addition. Following the strategy proposed in a previous low-T oxidation model of MCH [61], the rate constant of third O_2 addition was referred to that of second O_2 addition in the present model. In addition, considering the conformational effects, three main keto-dihydroperoxides (KDHP) formation pathways were proposed, as can be seen in Scheme 3. Their rate constants were referred to those of KHP formation pathways.

4. Results and discussion

4.1. Measured and simulated results

In this work, dozens of stable products and intermediates were observed in the ECH oxidation, including a variety of hydroperoxides like H_2O_2, CH_3OOH, C_2H_5OOH, $C_8H_{14}O_2$ (olefinic hydroperoxide - OFHP), $C_8H_{15}OOH$, $C_8H_{14}O_3$(KHP and CEHP), $C_8H_{14}O_4$ (olefinic dihydroperoxide - OFDHP) and $C_8H_{14}O_5$ (KDHP, cyclic ether dihydroperoxide - CEDHP). Figure 2 shows an example of mass spectrum which was recorded at 580 K (~30% fuel conversion) in the stoichiometric oxidation of ECH. Corresponding mass peaks of aforementioned species with m/z = 40–200 can be observed from the mass spectrum. The photoionization efficiency (PIE) spectra of some representative intermediates and their calculated IEs are presented in Figs. S6-S9 and Table S4 in the *Supplementary Material*, respectively. For small hydroperoxides like H_2O_2, CH_3OOH and C_2H_5OOH, the measured results are in good agreement with the IEs and PIE spectra in literature, as shown in Fig. S6. For large oxygenated intermediates, the observed onsets on the PIE spectra are in accordance with the lower boundaries of calculated IEs of possible structures. However, due to the enormous number of possible structures, it is not possible to distinguish every structure with unambiguous assignment to the observed ionization threshold. Thus in this work, the most representative structures of these large oxygenated intermediates were proposed based on co-evaluation of observed ionization thresholds and chemical nature.

Figures 3–5 present the measured mole fraction or normalized signal profiles of ECH, O_2, major products and intermediates. The measured and predicted results of ECH and O_2 are presented in Fig. 3(a,b). It can be observed from Fig. 3(a) that ECH presents remarkable NTC behaviors from lean to rich conditions, which can be captured by the present model. Similarly, the present model can well reproduce the mole fraction profiles of O_2 under the investigated conditions. It also shows generally reasonable performance in capturing the formation trends of major products and intermediates, except for several species under the lean condition. Furthermore, the present model was also validated against previous JSR oxidation data measured by GC–MS [19]. The results are shown in Fig. S10 in the *Supplementary Material*, from which the reasonable performance of the present model can also be observed.

In the low-T oxidation of alkanes, olefins and aldehydes/ketones are two kinds of typical intermediates produced from the decomposition of ROO and QOOH [10,11,47,62,78]. In this work, mass

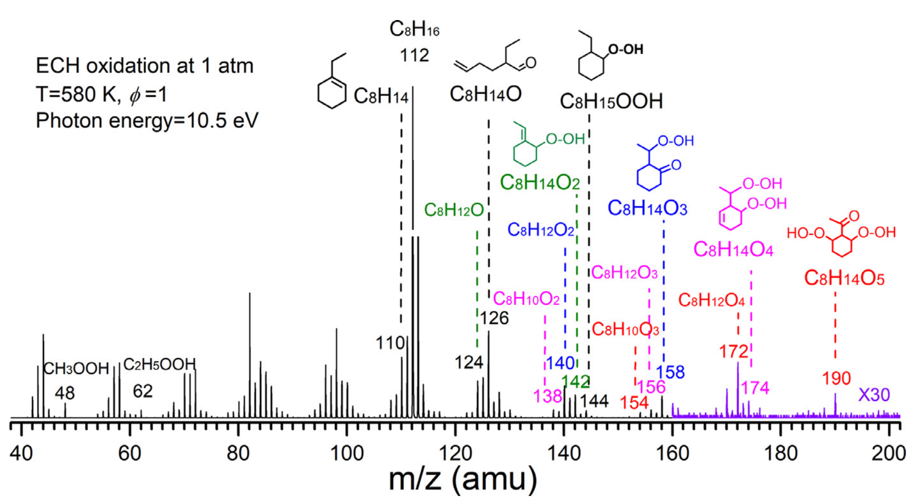


Fig. 2. Mass spectrum recorded at 10.5 eV, 580 K and $\phi = 1.0$. Important intermediates such as cycloalkene, alkenal, OFHP, ROOH, KHP, OFDHP, KDHP and their decomposition products are labeled with molecular weight, formula and most representative structures.

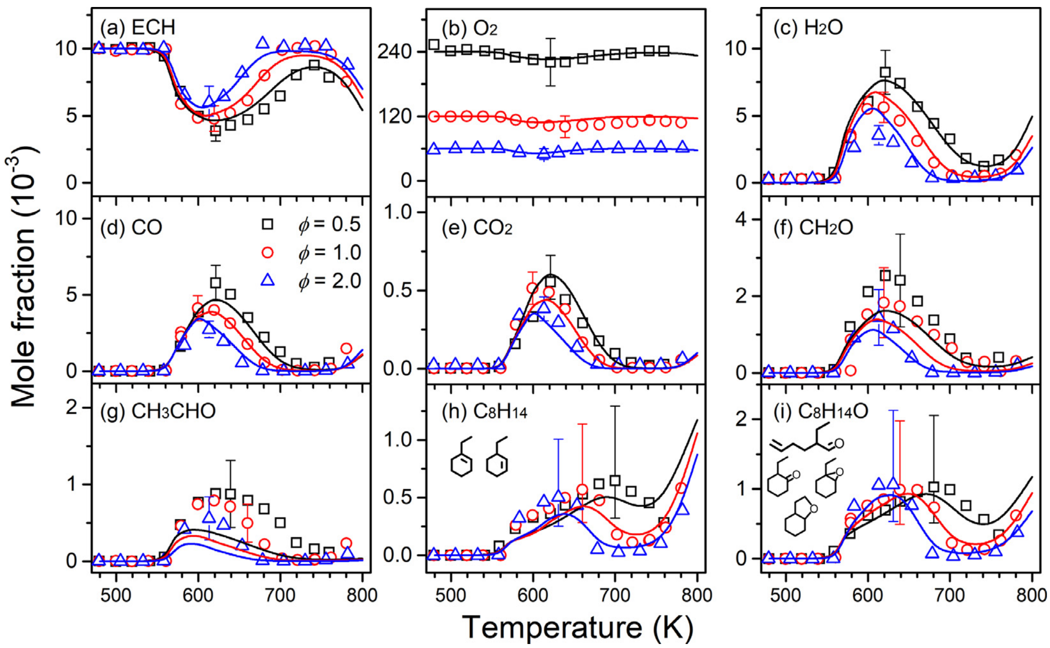


Fig. 3. Measured (symbols) and simulated (lines) mole fraction profiles of reactants, major products and intermediates at ϕ = 0.5, 1.0 and 2.0. Molecular structures in (h,i) belong to the most representative structures which also have highest simulated mole fractions among corresponding species.

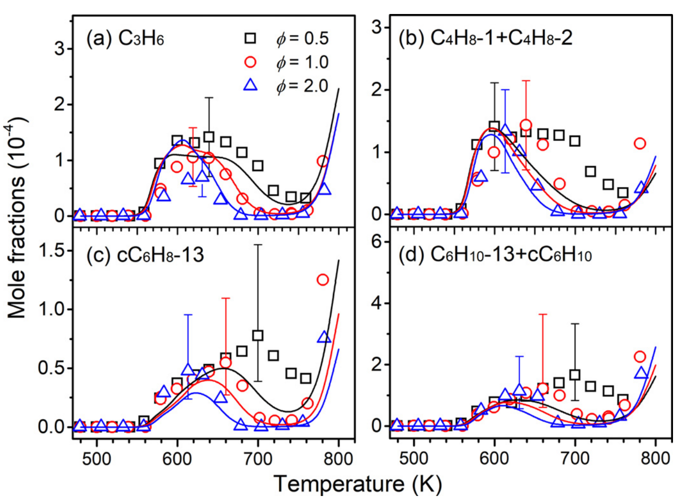


Fig. 4. Measured (symbols) and simulated (lines) mole fraction profiles of C_3–C_6 alkenes, dienes and cycloalkenes at ϕ = 0.5, 1.0 and 2.0.

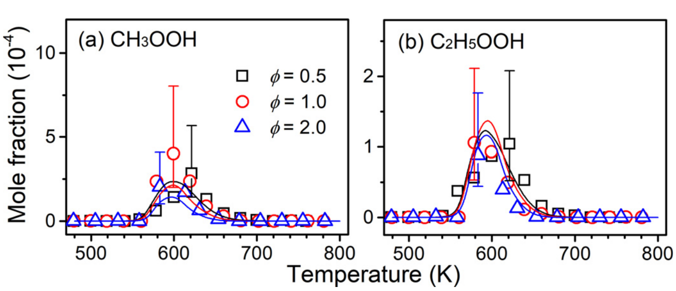


Fig. 5. Measured (symbols) and simulated mole fraction profiles (lines) of (a) methylhydroperoxide and (b) ethylhydroperoxide.

peaks at m/z = 110 and 126 were also observed, corresponding to the molecular weights of cycloalkenes (C_8H_{14}) and aldehydes/ketones with unsaturated carbon skeleton ($C_8H_{14}O$). C_8H_{14} has five possible cycloalkene isomers with literature IEs of 8.47–9.51 eV [79], while $C_8H_{14}O$ has more than ten possible isomers with calculated IEs of 8.95–9.72 eV, including alkenals, cyclic ethers and aldehydes/ketones with cyclic structures. As shown in Fig. S7(b,c), the onsets observed on PIE spectra of the two species are 8.45 eV and 8.95 eV respectively, which are close to the lower boundaries of literature or calculated IEs of C_8H_{14} and $C_8H_{14}O$ isomers. Figure 3(h,i) shows the total mole fraction profiles of these species at three equivalence ratios, while mole fraction profiles of separated isomers were achieved with GC–MS by Husson et al. [19], as shown in Fig. S10(f-l) in the *Supplementary Material*. As shown in Figs. 3 and S10, the present model can reasonably capture the mole fractions of C_8H_{14} and $C_8H_{14}O$ species at all equivalence ratios.

Compared with previous experimental studies of CHX, MCH and ECH using GC–MS [19,61,65], the measurements with SVUV-PIMS in this work provide additional information on a variety of hydroperoxides, as shown in Figs. 5 and 6. Since the PICSs of large hydroperoxides are not available, only normalized signal profiles are presented in Fig. 6. In this work, the signal profiles were normalized according to the simulated peak mole fraction values of corresponding species at ϕ = 1.0. As shown in Fig. 6(c), different from C_8H_{14} and $C_8H_{14}O$ which have wide temperature windows and peak at relatively high temperatures, $C_8H_{14}O_3$ species including KHPs, AnHPs and CEHPs are formed at very early stage (~550 K) and decay rapidly, which matches the first decomposition region of ECH oxidation shown in Fig. 3(a) very well. This phenomenon is in good accordance with the key role of KHPs in the low-T oxidation mechanism. Similar to KHPs, the temperature windows of CH_3OOH, C_2H_5OOH, $C_6H_{11}OOH$, $C_8H_{15}OOH$, $C_8H_{14}O_4$ and $C_8H_{14}O_5$ are also relatively narrow and can be captured by the present model. For $C_8H_{14}O_2$ in Fig. 6(d), the temperature window extends to the end of NTC region which can be captured by the simulated results of OFHP and its dione isomer together.

4.2. Chemistry in fuel decomposition and first O_2 addition

In order to reveal the key pathways in fuel decomposition and first O_2 addition processes, the ROP analysis was performed using the present model at different temperatures and ϕ = 1.0. Figure 7 shows the reaction network from ECH decomposition to the first

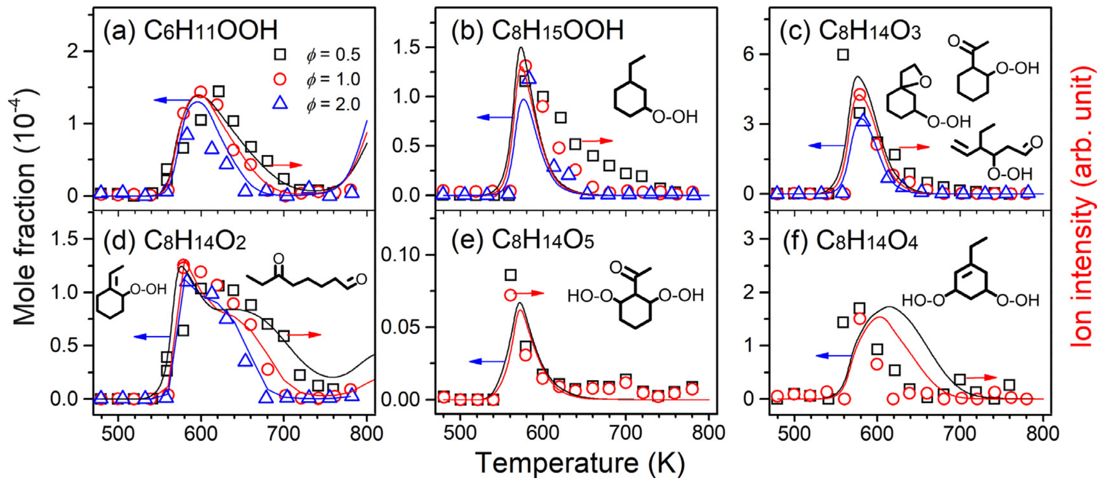


Fig. 6. Measured signal profiles (symbols) and simulated mole fraction profiles (lines) of (a) cyclohexyl hydroperoxide, (b) ROOH, (c) KHP/AnHP/CEHP, (d) OFHP, (e) KDHP/CEDHP and (f) OFDHP at 10.0 eV. Molecular structures belong to the most representative structures. The measured signal profile in each figure is normalized for a better illustration.

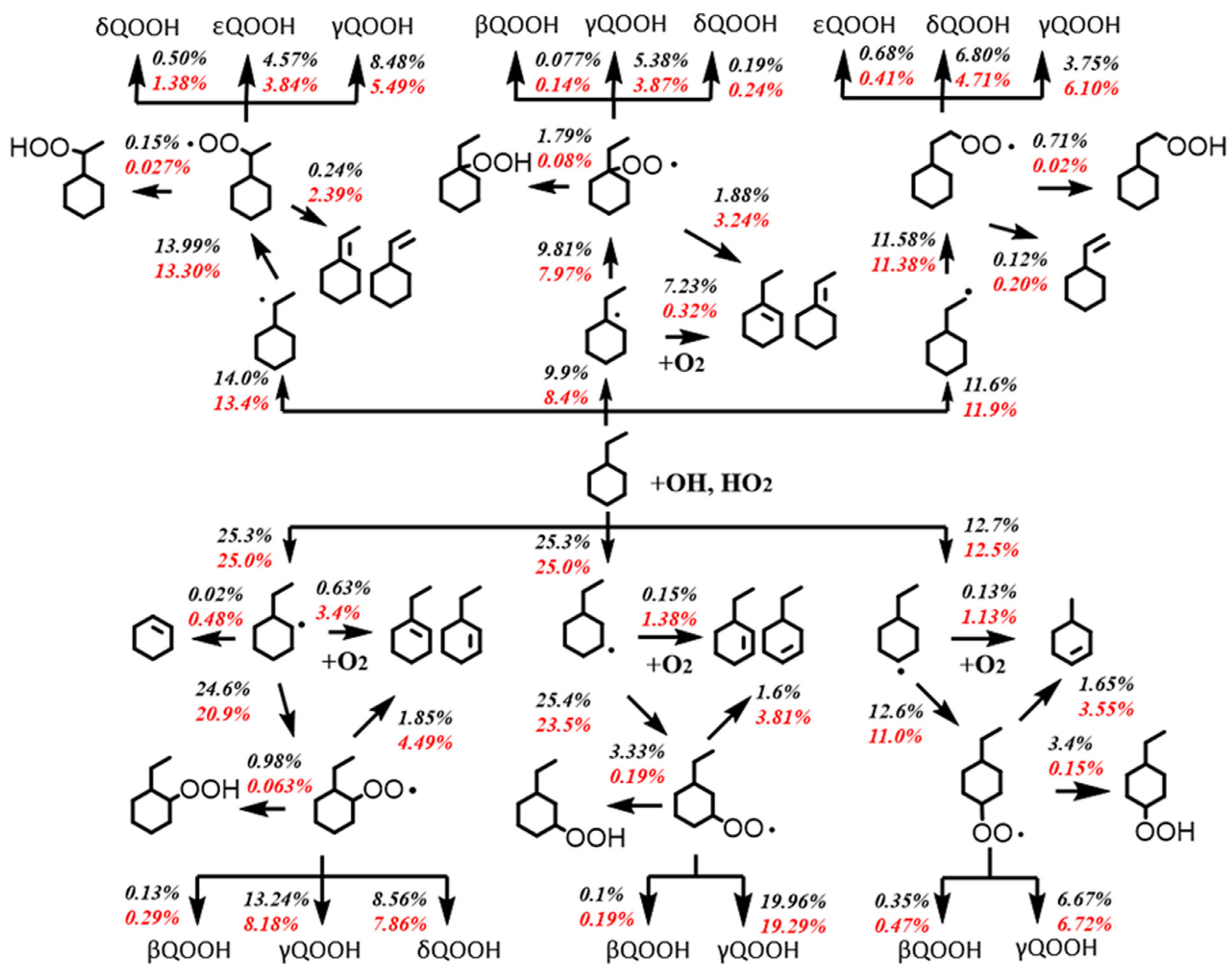


Fig. 7. Reaction network from ECH decomposition, first O_2 addition to ROO isomerization at two temperatures and $\phi = 1.0$. The reaction fluxes of corresponding pathway normalized by the total fluxes from ECH at 580 K and 680 K are highlighted in black and red, respectively. βQOOH, γQOOH, δQOOH and εQOOH denote the QOOHs formed via 5-, 6-, 7-m and 8-m TS rings, respectively.

O_2 addition under stoichiometric conditions. It can be observed that ECH is almost totally consumed via H-abstraction reactions to produce six fuel radicals under the investigated conditions. OH is the dominant radical in the H-abstraction of ECH, while HO_2 plays a minor role with less than 5% contributions below 800 K.

For all R radicals, the first O_2 addition reactions forming corresponding ROO radicals are their most important consumption pathways, while the “formally direct” chemically activated reactions of R + O_2 producing cycloalkenes+HO_2 has minor contributions. There are four consumption pathways of ROO, i.e.

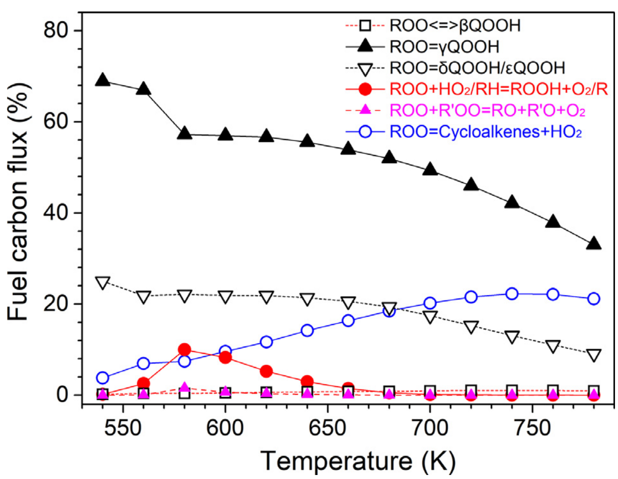


Fig. 8. Comparison among the fuel carbon fluxes of main ROO consumption pathways at 540–780 K and $\phi = 1.0$.

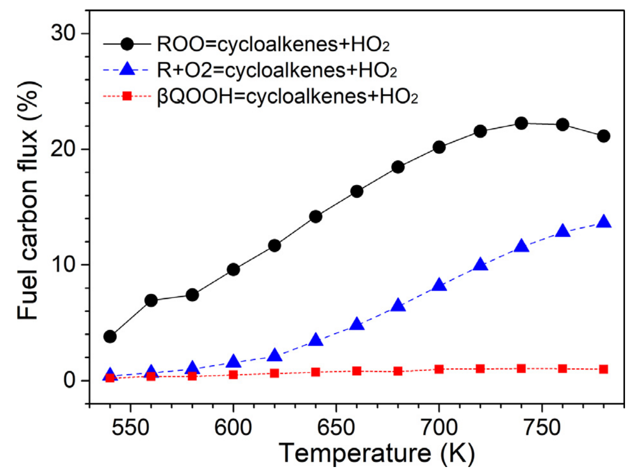


Fig. 9. Comparison of three main formation pathways of cycloalkenes at 540–780 K and $\phi = 1.0$.

isomerization to QOOH, concerted elimination to produce cycloalkenes+HO_2, reactions with HO_2 or fuel to produce ROOH and reactions with peroxy radicals to produce RO radicals. Figure 8 shows the contributions of the four pathways over 540–780 K at $\phi = 1.0$ based on the ROP analysis. At lower temperatures, isomerization to QOOH controls the consumption of ROO, while its contribution drops quickly as temperature increases. Among ROO isomerization reactions, the one via a 6-m TS ring has the largest contribution due to its lowest ring strain energy, while those via 7-m and 8-m TS rings also play important roles in ROO consumption. As temperature increases, concerted elimination becomes more and more important. Furthermore, the reactions of ROO with HO_2 or fuel are the dominant formation pathway of ROOH. At 580 K, about 10% of fuel is consumed via this pathway, indicating its importance in the first decomposition region of ECH oxidation. Its contribution decreases quickly with increasing temperature, which leads to the quick decay of ROOH at higher temperatures.

The decomposition of QOOH formed via different sizes of TS rings can produce cyclic ether/alkenal ($C_8H_{14}O$)+OH and serve as a typical chain-propagation step, while the decomposition of ROO and QOOH can produce cycloalkene (C_8H_{14})+HO_2 and serve as a chain-inhibition step in low-T oxidation, since HO_2 favors the self-combination reaction to yield H_2O_2 which is thermally stable below 800 K. In previous engine combustion studies, Yang et al. [27,80,81] and Kang et al. [20] found that cycloalkenes are a major product family in the low-T oxidation of alkylcyclohexanes. Previous studies [2,24] reveal that the cycloalkenes mainly produced by three pathways, including "formally direct" chemically activated reactions of R + O_2, concerted elimination reactions of ROO and β-C-O scission reactions of βQOOH. It should be noted that there are still debates on the dominant formation pathway of cycloalkenes in the low-T oxidation of cycloalkanes, especially the first two pathways. For example, Kang et al. [20] proposed that alkylcycloalkenes were mainly formed through the 1,4-H shift pathway based on the conformational analysis. Besides, Knepp et al. [2] and Fernandes et al. [24] suggested that the "formally direct" chemically activated reaction of R + O_2 greatly contributes to cycloalkene formation in CHX oxidation. In the present model, all of the three pathways were incorporated. Figure 9 shows their contributions to the ECH consumption over 540–780 K at $\phi = 1.0$ based on the ROP analysis. The concerted elimination of ROO is the most important formation pathway of cycloalkenes, while the β-C-O scission of βQOOH has much lower contributions. It should be noted that "formally direct" chemically activated reactions of R + O_2 also have remarkable contributions to the formation of cycloalkenes, especially at temperatures higher than 700 K. As a result, the ROP analysis in this work demonstrates that both the concerted elimination reactions of ROO and "formally direct" chemically activated reactions of R + O_2 have important contributions to the formation of cycloalkenes in low-T oxidation of ECH.

4.3. Chemistry in chain-branching process and NTC behavior

To reveal the driving forces in determining the reactivity of ECH oxidation at different temperature regions, i.e. the first decomposition (540–620 K), NTC (620–740 K) and second decomposition (740–780 K) regions, the ROP analysis and sensitivity analysis were performed under different temperatures and $\phi = 1.0$. Since there are six ROO radicals, the ECHR2OO radical which can have further reactions on both ring and sidechain carbon sites is selected as the representative one and its further reaction network is illustrated in Fig. 10. A simplified reaction scheme of low-T oxidation of ECH covering the three temperature regions is summarized in Scheme 4.

Low-temperature oxidation region: As shown in Scheme 4(a), similar to alkanes [47,62] and other cycloalkanes [30,61], the main chain-branching reaction sequence in the first decomposition region of ECH oxidation includes: (1) more than 95% R undergoes the first O_2 addition; (2) more than 53% ROO isomerizes to γQOOH through a 6-m TS ring; (3) γQOOH undergoes the second O_2 addition to form OOγQOOH; (4) OOγQOOH decomposes to KHP and the first OH radical; (5) KHP decomposes to XO and the second OH radical. Therefore, the following reaction sequences

$$\mathrm{ECHR2OO} \leftrightarrow \mathrm{ECH2Q7J} \overset{O2}{\Leftrightarrow} \mathrm{ECH2Q7QJ} \rightarrow \mathrm{ECH2O7Q} + \mathrm{OH} \rightarrow \mathrm{ECH2O7OJ} + 2\mathrm{OH} \quad (1)$$

$$\mathrm{ECHR7OO} \leftrightarrow \mathrm{ECH7Q2J} \overset{O2}{\Leftrightarrow} \mathrm{ECH7Q2QJ} \rightarrow \mathrm{ECH7O2Q} + \mathrm{OH} \rightarrow \mathrm{ECH7O2OJ} + 2\mathrm{OH} \quad (2)$$

become the main chain-branching reaction sequences in low-T oxidation of ECH. Besides the KHP pathway, there are three other chain-branching pathways via AnHP, KDHP and ROOH, as shown in Scheme 4(a). Figure 11 compares their contributions to chain branching over 540–780 K at $\phi = 1.0$. It can be observed that the KHP pathway dominates the OH formation over the whole temperature region. The ROOH pathway contributes secondly over 600–700 K, while the newly proposed AnHP

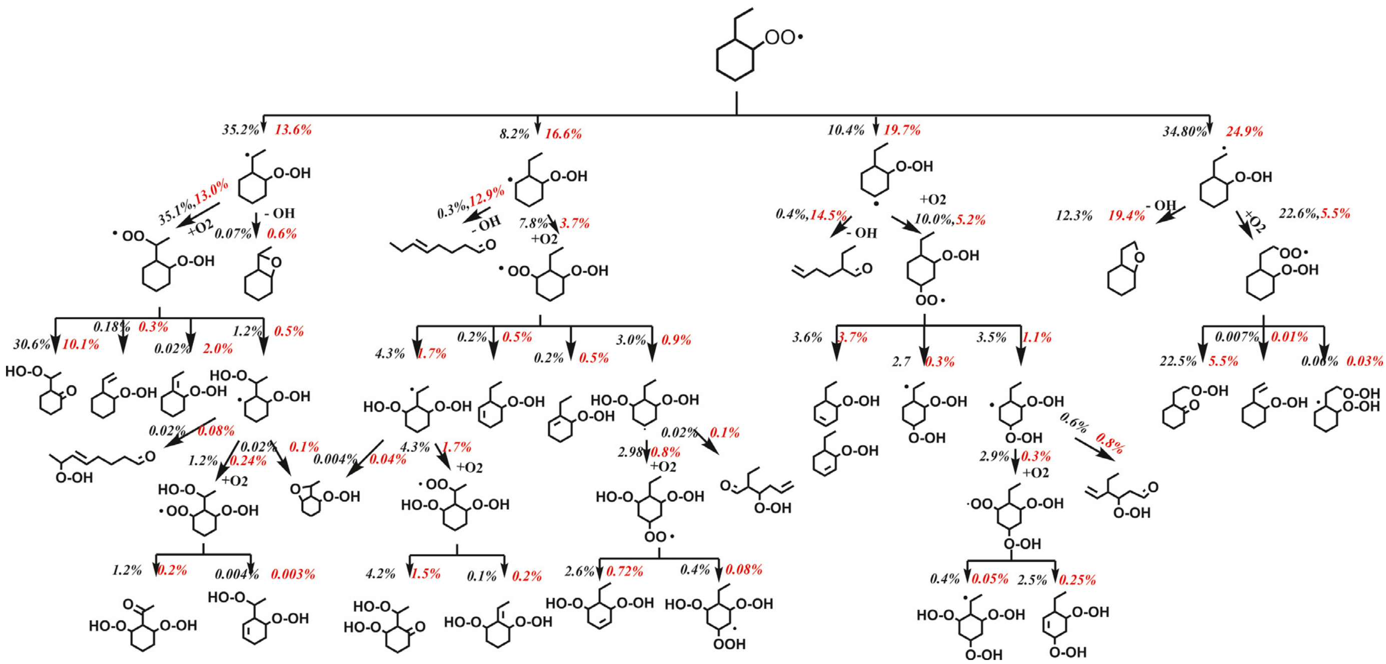
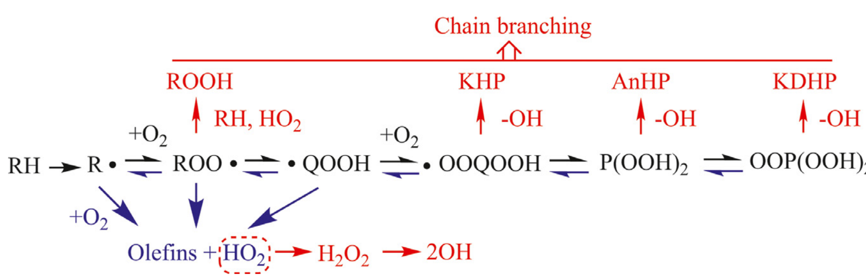


Fig. 10. Reaction network from ECHR2OO radical at two temperatures and ϕ = 1.0. The reaction fluxes of corresponding pathway normalized by the total fluxes from ECH at 580 K and 680 K are highlighted in black and red, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



Scheme 4. Simplified reaction scheme in low-T oxidation of ECH: (a) first decomposition region, (b) NTC region and (c) second decomposition region.

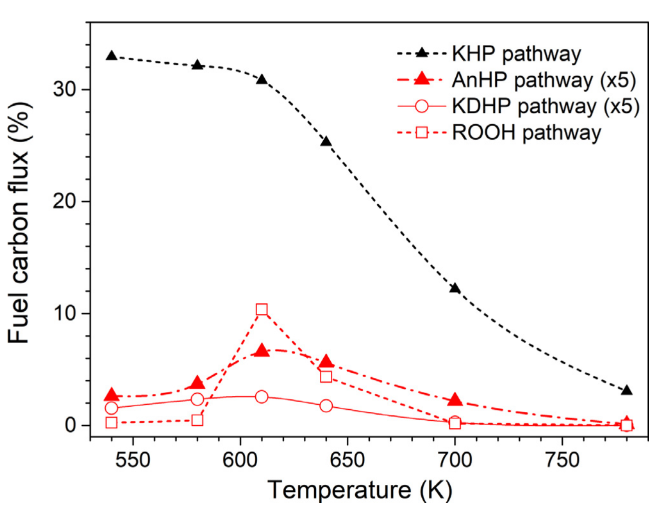


Fig. 11. Comparison of four chain-branching pathways via KHP, AnHP, KDHP and ROOH at 540–780 K and ϕ = 1.0.

pathway has similar contributions to the KDHP pathway. It should be mentioned that for ECHR3OO which can only undergo 1,5-H shift on the ring carbon sites, the contributions of AnHP pathway to ECHR3OO decomposition increase dramatically to 10.0%. This implies that the AnHP pathway can very likely play a much more important role in low-T oxidation of cycloalkanes without alkyl sidechains.

Besides, Fig. 12(b) presents the sensitivity analysis of ECH at ϕ = 1.0 and T = 580, 680 and 780 K. Three typical fuel decomposition ($ECH+OH=ECHR7+H_2O$) and second O_2 addition ($ECH2QJ7+O_2=ECH2Q7QJ$ and $ECH7QJ2+O_2=ECH7Q2QJ$) reactions become the most sensitive reactions to ECH consumption at 580 K since the three reactions finally contribute to the formation of key KHPs (K7HP2 and K2HP7). Furthermore, it is also worth mentioning that beside the 6-m TS ring pathway forming γQOOH, the ECHR2OO isomerization pathway via a 7-m TS ring forming δQOOH (ECH2QJ8) contributes 22.5% to ECHR2OO consumption at 580 K and finally results in the formation of another KHP (K2HP8) and OH. The reaction flux of the 7-m TS ring pathway is almost half of that of the 6-m TS ring pathway. Therefore, the 7-m TS ring pathway also has a remarkable contribution to the fuel consumption and chain branching in the first decomposition region of ECH oxidation.

Besides the conventional two-stage O_2 addition mechanism, OOγQOOH can also isomerize to $\gamma P(OOH)_2$ via 1,5-H shift, which initiates the third O_2 addition mechanism. Finally, $OO\gamma P(OOH)_2$ mainly decomposes to HOMs, including KDHP ($C_8H_{14}O_5$) and OFDHP ($C_8H_{14}O_4$). Further decomposition of KDHP and OFDHP can produce X′O radical and release two OH radicals. However,

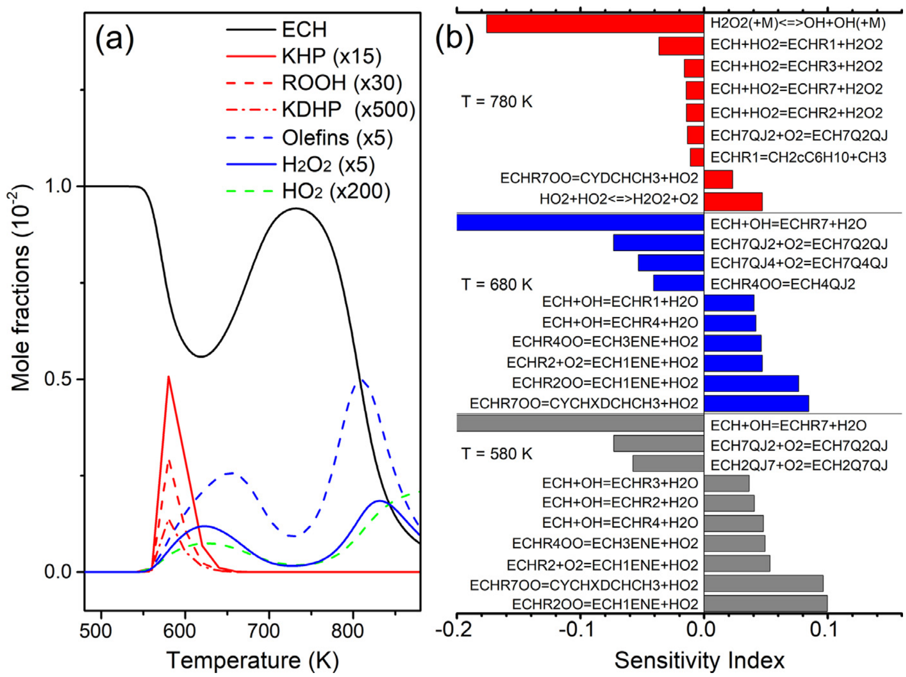


Fig. 12. (a) Predicted mole fractions of ECH and chain-branching tracers (ROOH, KHP and KDHP) and chain-termination tracers (olefins, H_2O_2 and HO_2) at 480–900 K and ϕ = 1.0. (b) Sensitivity analysis of ECH at 580, 680 and 780 K and ϕ = 1.0.

as shown in Fig. 6, the predicted peak mole fractions of these HOMs are around one order of magnitude lower than those of KHP/AnHP/CEHP ($C_8H_{14}O_3$) and OFHP ($C_8H_{14}O_2$). Furthermore, the ROP analysis in Fig. 10 shows that only 8.5% of ECHR2OO eventually converts to HOMs, while the percentage is 53.1% for KHPs. It can be concluded that in low-T oxidation of ECH at atmospheric pressure, the contribution of third O_2 addition to chain branching can hardly be competitive to that of second O_2 addition.

NTC region: As temperature increases, the main chain-branching reaction sequence in the first decomposition region is inhibited. At 680 K, the ROP analysis in Fig. 10 shows that the reaction flux from ECHR2OO to KHPs decreases to 15.6%. Furthermore, the contribution of third O_2 addition mechanism drops from 8.5% to 2.1%. On the contrary, the reverse reactions of first O_2 addition and cycloalkene formation reactions become much more active. As a result, the concentration of OH which is the dominant chain carrier in low-T oxidation decreases rapidly. Meanwhile, the total contribution of cycloalkenes and HO_2 formation pathways increases dramatically with increasing temperature at T < 740 K, as shown in Fig. 9 and Scheme 4(b). The sensitivity analysis results shown in Fig. 12(b) demonstrates that cycloalkenes and HO_2 formation pathways present high sensitivity coefficients to fuel consumption in this region. In particular, the self-combination of HO_2 producing H_2O_2 and O_2 enhances the chain termination. Besides, the ROP analysis shows that the reaction flux from ECHR2OO to alkenal and OH also increases from 0.7% to 27.4%. Consequently, the low-T oxidation reactivity of ECH is inhibited, leading to the apparent NTC behavior in ECH oxidation.

High-T oxidation region: When temperature is higher than 740 K, the formation of HO_2 keeps on increasing, as shown in Fig. 12(a). Consequently ECH+HO_2=R + H_2O_2 becomes more and more important and serves as an important consumption pathway of ECH. Fig. 12(a) also shows that the H_2O_2 is gradually accumulated as the concentration of HO_2 increases. When temperature exceeds 830 K, H_2O_2 becomes thermally instable and decomposes to two OH radicals, as can be seen from Scheme 4(c). This reaction serves as the new chain-branching pathway and further oxidation of ECH is triggered. The sensitivity analysis in Fig. 12(b) also verifies the importance of ECH+HO_2=R + H_2O_2 and H_2O_2==OH+OH in enhancing the oxidation reactivity of ECH at 780 K.

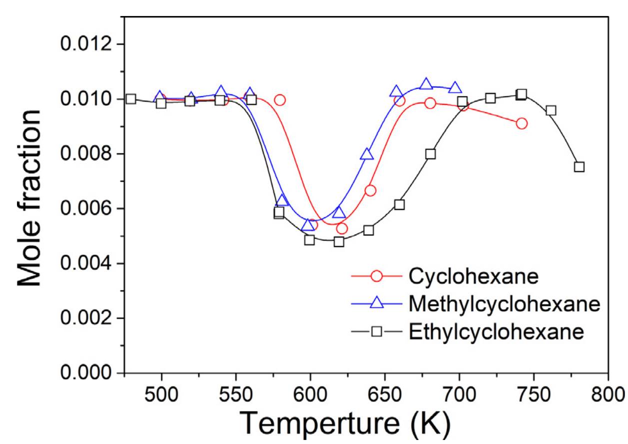
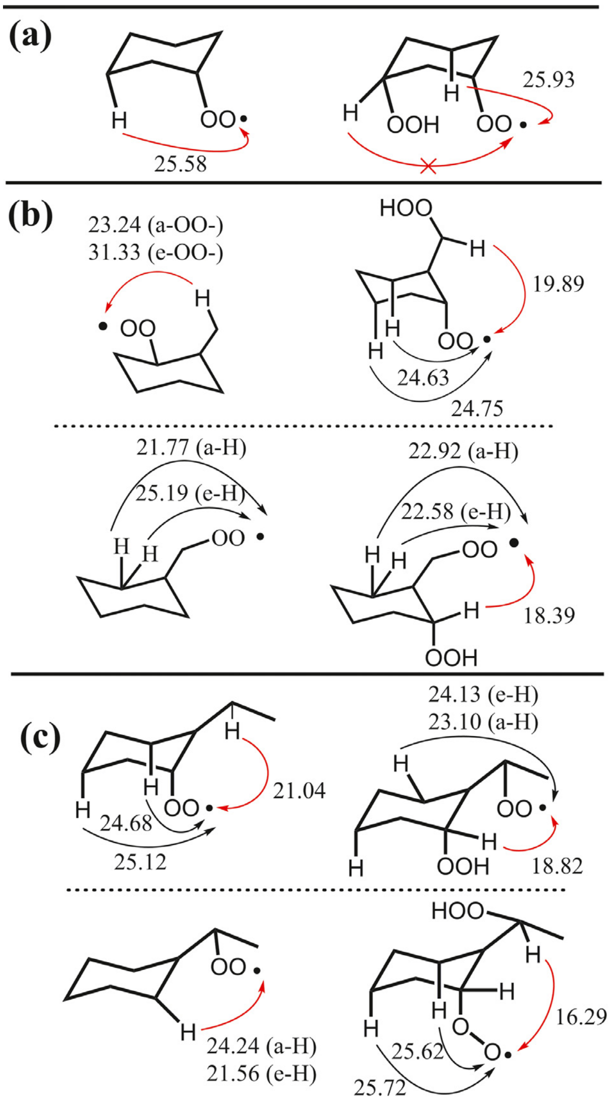


Fig. 13. Measured mole fraction profiles of fuels at ϕ = 1 and residence time of 2 s in low-T oxidation of CHX [82], MCH [61] and ECH at pressures close to atmospheric pressure.

4.4. *Effects of ethyl sidechain on the low-T oxidation reactivity*

As shown in Fig. 1, isomerization reactions such as ROO=QOOH and OOQOOH=KHP+OH play crucial roles in the low-T chain-branching process. In particular, those covering both the sidechain and ring carbon sites are very influential based on the ROP analysis. Therefore, in this section, the effect of ethyl sidechain on the low-T oxidation reactivity is explored based on both modeling and conformational analyses.

As shown in Fig. 13, ECH shows more profound low-T oxidation reactivity than CHX [82] and MCH [61]. The window of low-T ECH oxidation starts from 560 K and ends at 700 K, while the windows of CHX and MCH locate at 580–660 K and 560–650 K, respectively. Thus the low-T oxidation reactivity of the three fuels follows the trend of CHX<MCH<ECH, which indicates that the



Scheme 5. Conformational analysis of ROO and OOγQOOH isomerization in low-T oxidation of (a) CHX, (b) MCH and (c) ECH which is related to KHP formation. The numbers are energy barriers of corresponding pathways with the unit of kcal/mol.

presence and increased length of alkyl sidechain can increase the low-T reactivity of cycloalkanes. The enhanced reactivity of MCH and ECH compared with that of CHX is mainly due to the existence of alkyl sidechains. The enhanced reactivity of ECH compared with that of MCH is mainly due to the high rotational freedom of ethyl sidechain and one more secondary carbon site on the sidechain.

To further explore the sidechain effects, Scheme 5 compares the energy barriers of the isomerization reactions of typical ROO and OOQOOH in low-T oxidation of CHX, MCH and ECH. As discussed above, the cyclohexane ring inhibits the formation of KHP from OOQOOH with both OO• and OOH moieties on the ring carbon sites, as shown in Scheme 5(a). Thus CHX has the weakest low-T chain-branching process and lowest low-T reactivity among all three cycloalkanes. In low-T oxidation of MCH and ECH, as shown in Scheme 5(b,c), the high rotational freedom of alkyl sidechain provides great convenience for the ROO and OOQOOH isomerization processes, which can reduce the energy barrier in ROO isomerization process and facilitate KHP formation compared with low-T oxidation of CHX. As a result, the low-T oxidation reactivity of MCH and ECH was promoted. Furthermore, the energy barriers of KHP formation reactions in low-T oxidation of MCH are 1–2 kcal/mol higher than similar reactions in low-T oxidation of ECH due to stronger C–H bonds in methyl sidechain. Therefore, ECH demonstrates the highest low-T reactivity and broadest low-T oxidation window, as shown in Fig. 13.

5. Conclusions

In this work, the low-T oxidation of ECH was performed in a JSR at 780 Torr, 480–780 K and equivalence ratios form 0.5 to 2.0. IEs of some important intermediates and energy barriers of several important pathways were theoretically calculated at CBS-QB3 level. A low-T ECH oxidation model, incorporating the high-T oxidation reaction classes, conventional low-T reaction classes, third O_2 addition pathways and some other additional pathways related to low-T oxidation chemistry, was constructed in this work. The major conclusions are summarized below.

(1) The application of SVUV-PIMS facilitates the identification and quantification of oxidation products, especially many reactive intermediates. The calculated IEs were used to assist the identification of some important intermediates. Due to the complex isomer distributions for large oxygenated species, it is difficult to obtain unambiguous isomeric identification for them, while the observed onsets on the PIE spectra are in accordance with the lower boundaries of calculated IEs of possible structures. In general, C_8H_{14} and $C_8H_{14}O$ were observed as the most abundant large intermediates, while a series of hydroperoxides were also detected.

(2) The present model can reasonably capture the low-temperature oxidation reactivities and negative temperature coefficient behaviors observed in both present and previous experimental work of ethylcyclohexane oxidation. Based on conformational analysis, AnHP pathway was proposed for the isomerization of OOQOOH with both OO• and OOH moieties on the ring carbon sites, while this pathway is found very likely to play a much more important role in the low-T oxidation of cycloalkanes without alkyl sidechains. In the future, low-T IDT data measured in rapid compression machine are desired for further validation of the present model, especially under high pressure conditions.

(3) Based on modeling analyses, the two-stage O_2 addition mechanism is concluded to dominate the chain-branching process in the low-temperature region, mainly through reaction sequences R + O_2=ROO=QOOH, QOOH+O_2==OOQOOH=KHP+OH and KHP=XO+OH, which contribute the most to OH formation. Compared with the second O_2 addition, the third O_2 addition only presents limited contribution to chain branching. The concerted elimination of ROO and "formally direct" chemically activated reactions of R + O_2 have important contributions to cycloalkenes and HO_2 formation and serve as main chain-termination pathways.

(4) The alkyl sidechain has great influence on the low-T oxidation reactivity of cycloalkanes due to its high rotational freedom. Compared with smaller cycloalkanes like CHX and MCH, the ethyl sidechain structure in ECH reduces the energy barriers of ROO isomerization and facilitates the formation of KHP, which leads to more pronounced low-temperature oxidation reactivity of ECH.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.12.038.

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